

2.7**Solving Nonlinear Systems of Equations**

For use with Exploration 2.7

Essential Question How can you solve a system of two equations when one is linear and the other is quadratic?

1 EXPLORATION: Solving a System of Equations

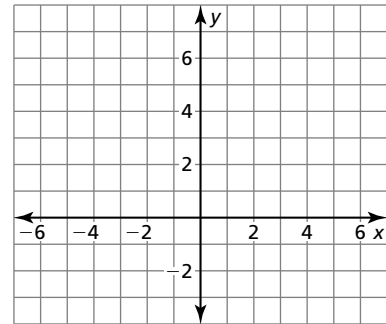
Go to *BigIdeasMath.com* for an interactive tool to investigate this exploration.

Work with a partner. Solve the system of equations by graphing each equation and finding the points of intersection.

System of Equations

$$y = x + 2 \quad \text{Linear}$$

$$y = x^2 + 2x \quad \text{Quadratic}$$

**2 EXPLORATION:** Analyzing Systems of Equations

Work with a partner. Match each system of equations with its graph (shown on the next page). Then solve the system of equations.

a. $y = x^2 - 4$
 $y = -x - 2$

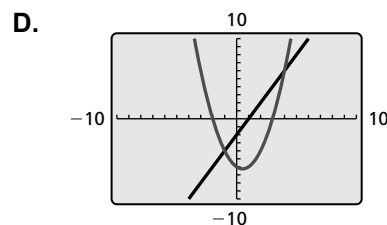
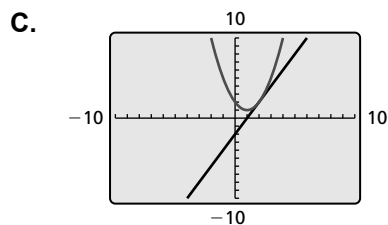
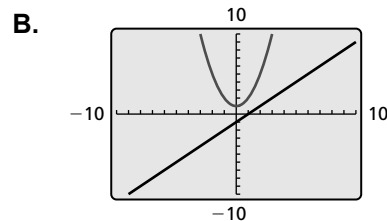
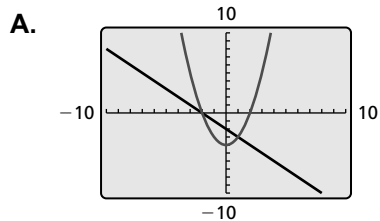
b. $y = x^2 - 2x + 2$
 $y = 2x - 2$

c. $y = x^2 + 1$
 $y = x - 1$

d. $y = x^2 - x - 6$
 $y = 2x - 2$

2.7 Solving Nonlinear Systems of Equations (continued)

2 EXPLORATION: Analyzing Systems of Equations (continued)



Communicate Your Answer

- How can you solve a system of two equations when one is linear and the other is quadratic?
- Write a system of equations (one linear and one quadratic) that has (a) no solutions, (b) one solution, and (c) two solutions. Your systems should be different from those in Explorations 1 and 2.

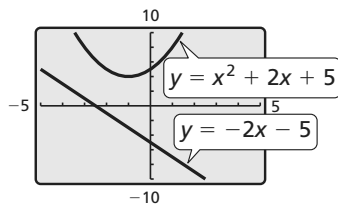
2.7**Practice**

For use after Lesson 2.7

Notes:**Worked-Out Examples****Example #1****Solve the system by graphing.**

$$y = x^2 + 2x + 5$$

$$y = -2x - 5$$



The graphs do not intersect. So, the system has no solutions.

Example #2**Solve the system by substitution.**

$$y = -x^2 + 7$$

$$y = 2x + 4$$

Step 2 $y = -x^2 + 7$

$$2x + 4 = -x^2 + 7$$

$$2x + 4 + x^2 - 7 = -x^2 + x^2 + 7 - 7$$

$$x^2 + 2x - 3 = 0$$

$$(x + 3)(x - 1) = 0$$

$$x + 3 = 0 \quad \text{or} \quad x - 1 = 0$$

$$\frac{-3}{x} = \frac{-3}{-3} \quad \frac{+1}{x} = \frac{+1}{1}$$

Step 3 $y = 2x + 4$

$$y = 2(-3) + 4$$

$$y = -6 + 4$$

$$y = -2$$

$$y = 2x + 4$$

$$y = 2(1) + 4$$

$$y = 2 + 4$$

$$y = 6$$

The solutions are $(-3, -2)$ and $(1, 6)$.

2.7 Practice (continued)**Example #3**

Solve the system by elimination.

$$y = 2x - 1$$

$$y = x^2$$

$$\begin{array}{r} \text{Step 2} \quad y = x^2 \\ -(y = 2x - 1) \\ \hline 0 = x^2 - 2x + 1 \end{array}$$

$$\begin{array}{r} \text{Step 3} \quad 0 = x^2 - 2(x)(1) + 1^2 \\ 0 = (x - 1)^2 \\ x - 1 = 0 \\ \quad \quad \quad \frac{+1}{+1} \quad \frac{+1}{+1} \\ \quad \quad \quad \hline \quad \quad \quad x = 1 \end{array}$$

$$\begin{array}{r} \text{Step 4} \quad y = x^2 \\ y = 1^2 \\ y = 1 \end{array}$$

The solution is (1, 1).

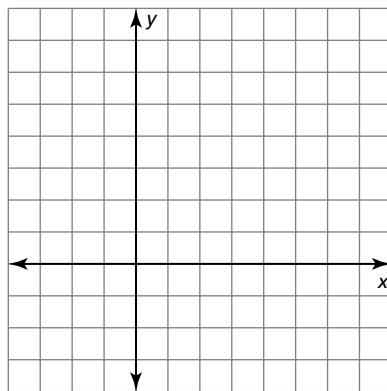
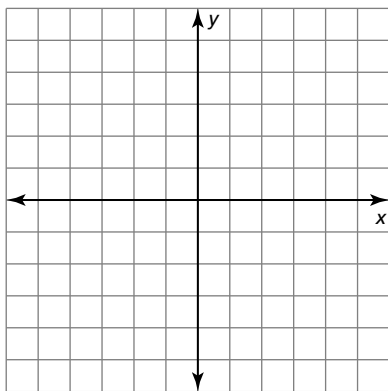
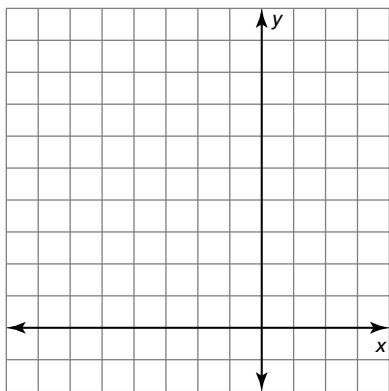
Practice A

In Exercises 1–6, solve the system by graphing.

$$\begin{array}{l} 1. \quad y = x^2 + 5x + 6 \\ \quad y = -x + 1 \end{array}$$

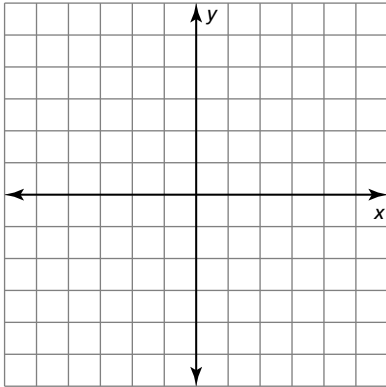
$$\begin{array}{l} 2. \quad y = x^2 + x - 3 \\ \quad y = x + 1 \end{array}$$

$$\begin{array}{l} 3. \quad y = \frac{1}{2}x^2 - 2x + 1 \\ \quad y = -x + 1 \end{array}$$

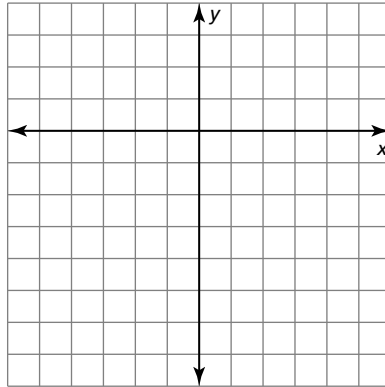


2.7 Practice (continued)

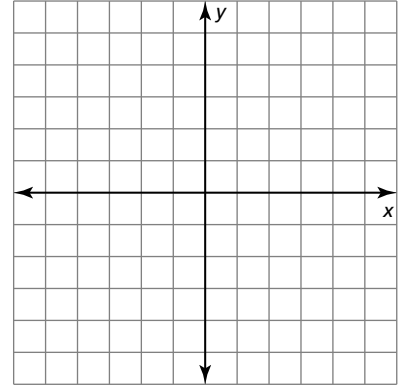
4. $y = -3x^2 - 3x + 2$
 $y = 2x$



5. $y = -\frac{1}{3}x^2 + x - 2$
 $y = -2$



6. $y = 6x^2 + 3x - 5$
 $y = -3x - 5$



In Exercises 7–9, solve the equation by substitution.

7. $y - 2 = x^2$
 $y = 6$

8. $y = -2x^2$
 $y = 3x + 2$

9. $y = x - 4$
 $y = x^2 + 3x - 4$

In Exercises 10–12, solve the equation by elimination.

10. $y = x^2$
 $y = x - 3$

11. $y = x^2 + 3x - 5$
 $y = 3x - 1$

12. $y = x^2 + x - 2$
 $y = x + 14$

In Exercises 13–18, solve the equation. Round your solution(s) to the nearest hundredth, if necessary.

13. $-6x + 14 = x^2 - 9x + 16$

14. $-x^2 + 4x = -2x + 8$

2.7 Practice (continued)

15. $4x^2 - 9 = 4x - 1$

16. $-\frac{1}{2}x + 1 = -x^2 + 4x$

17. $2x^2 - 4 = -x^2 + 6$

18. $-3\left(\frac{2}{3}\right)^x + 2 = x^2 - 2$

Practice B**In Exercises 1–4, solve the system by graphing.**

1. $y = 4x^2 + 2x - 1$
 $y = -2x + 7$

2. $y = \frac{1}{3}x^2 - 6x + 5$
 $y = -5x + 5$

3. $y = 4x^2 - 8x$
 $y = -4$

4. $y = 3x^2 - 2x + 8$
 $y = -x$

In Exercises 5–8, solve the system by substitution.

5. $y = 6x$
 $y = x^2 + 9$

6. $y = 2x - 5$
 $y = 2x^2 - 3x + 3$

7. $y = -x^2 - 2x + 4$
 $y = 3x - 10$

8. $y + 3 = x^2$
 $y = -3$

In Exercises 9–12, solve the system by elimination.

9. $y = x^2 - x - 1$
 $y = x - 2$

10. $y = 2x^2 + 2x$
 $y = -2x + 6$

11. $y = x^2 - 4x + 7$
 $y = -x + 11$

12. $y = -x^2 + 1$
 $y = 2x - 2$

In Exercises 13 and 14, use the table to describe the location of the zeros of the quadratic function f .

13.

x	-2	-1	0	1	2	3
f(x)	-3	-2	-2	0	1	2

14.

x	-1	0	1	2	3	4
f(x)	3	2	-1	2	3	5

15. The graphs of $f(x) = 1.6x^2 + 2x - 0.6$ and $g(x) = -2.5x^2 - 2x - 4.2$ do not intersect. Change the value(s) of c in one or both functions f and g until the two graphs do intersect. Write your new system of equations and determine the intersection point(s), rounding to the nearest hundredth if necessary.